

# District 5 SPSS Comparison 2023-2025 Sales

Elliot Dean - Program Analyst IV  
Office of the Chief Financial Officer | Office of the Assessor  
City of Detroit

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## Contents

|                     |   |
|---------------------|---|
| Model I . . . . .   | 1 |
| Model II . . . . .  | 1 |
| Model III . . . . . | 2 |
| Model IV . . . . .  | 2 |
| Model V . . . . .   | 2 |
| Model VI . . . . .  | 3 |
| Plots . . . . .     | 4 |

## Model I

Table 1: District 5 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 160      | 153      | 7      | 0.6275 | 0.3930 | 0.3650 | 14.1390 |
| R                  | 160      | 153      | 7      | 0.6275 | 0.3928 | 0.3650 | 14.1386 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.05% | 0.00%  | -0.00%  |

## Model II

Table 2: District 5 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 307      | 289      | 18     | 0.6744 | 0.5350 | 0.5060 | 18.4470 |
| R                  | 307      | 289      | 18     | 0.6744 | 0.5346 | 0.5057 | 18.4465 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.07% | -0.06% | -0.00%  |

## Model III

Table 3: District 5 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 307      | 281      | 26     | 0.5948 | 0.6480 | 0.6150 | 19.8960 |
| R                  | 307      | 281      | 26     | 0.5948 | 0.6480 | 0.6154 | 19.8957 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.00%  | 0.07%  | -0.00%  |

Table 4: District 5 | LN\_PRICE Model Assessing Statistics: SPSS vs R

| metricsource       | median_ratio | mean_ratio | wmean_ratio | PRD    | COD     | PRB     |
|--------------------|--------------|------------|-------------|--------|---------|---------|
| SPSS               | 1.0000       | 1.2930     | 0.9090      | 1.4230 | 0.6450  |         |
| R                  | 0.9803       | 1.1826     | 0.9003      | 1.3136 | 0.5156  | -0.2192 |
| % diff (R vs SPSS) | -1.97%       | -8.54%     | -0.96%      | -7.69% | -20.07% |         |

## Model IV

Table 5: District 5 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 262      | 238      | 24     | 0.4466 | 0.7790 | 0.7570 | 35.0370 |
| R                  | 260      | 237      | 23     | 0.4452 | 0.7767 | 0.7550 | 35.8318 |
| % diff (R vs SPSS) | -0.76%   | -0.42%   | -4.17% | -0.32% | -0.30% | -0.26% | 2.27%   |

Table 6: District 5 | LN\_PRICE (Inliers) Model Assessing Statistics: SPSS vs R

| metricsource       | median_ratio | mean_ratio | wmean_ratio | PRD    | COD     | PRB     |
|--------------------|--------------|------------|-------------|--------|---------|---------|
| SPSS               | 1            | 1.1470     | 0.9440      | 1.2150 | 0.4370  | -0.1290 |
| R                  | 1            | 1.0948     | 0.9384      | 1.1667 | 0.3630  | -0.1106 |
| % diff (R vs SPSS) | -0.00%       | -4.55%     | -0.60%      | -3.98% | -16.93% | 14.28%  |

## Model V

Table 7: District 5 | LN\_TASP Model I Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 262      | 239      | 23     | 0.4458 | 0.7740 | 0.7520 | 35.5210 |
| R                  | 260      | 236      | 24     | 0.4779 | 0.7437 | 0.7177 | 28.5403 |
| % diff (R vs SPSS) | -0.76%   | -1.26%   | 4.35%  | 7.21%  | -3.91% | -4.56% | -19.65% |

Table 8: District 5 | LN\_TASP Model I Assessing Statistics: SPSS vs R

| metricsource       | median_ratio | mean_ratio | wmean_ratio | PRD    | COD     | PRB     |
|--------------------|--------------|------------|-------------|--------|---------|---------|
| SPSS               | 1.0020       | 1.1460     | 0.9430      | 1.2150 | 0.4360  | -0.1300 |
| R                  | 1.0140       | 1.1053     | 0.9312      | 1.1870 | 0.3842  | -0.1346 |
| % diff (R vs SPSS) | 1.19%        | -3.55%     | -1.26%      | -2.30% | -11.89% | -3.54%  |

## Model VI

Table 9: District 5 | LN\_TASP Model II Fit Statistics: SPSS vs R

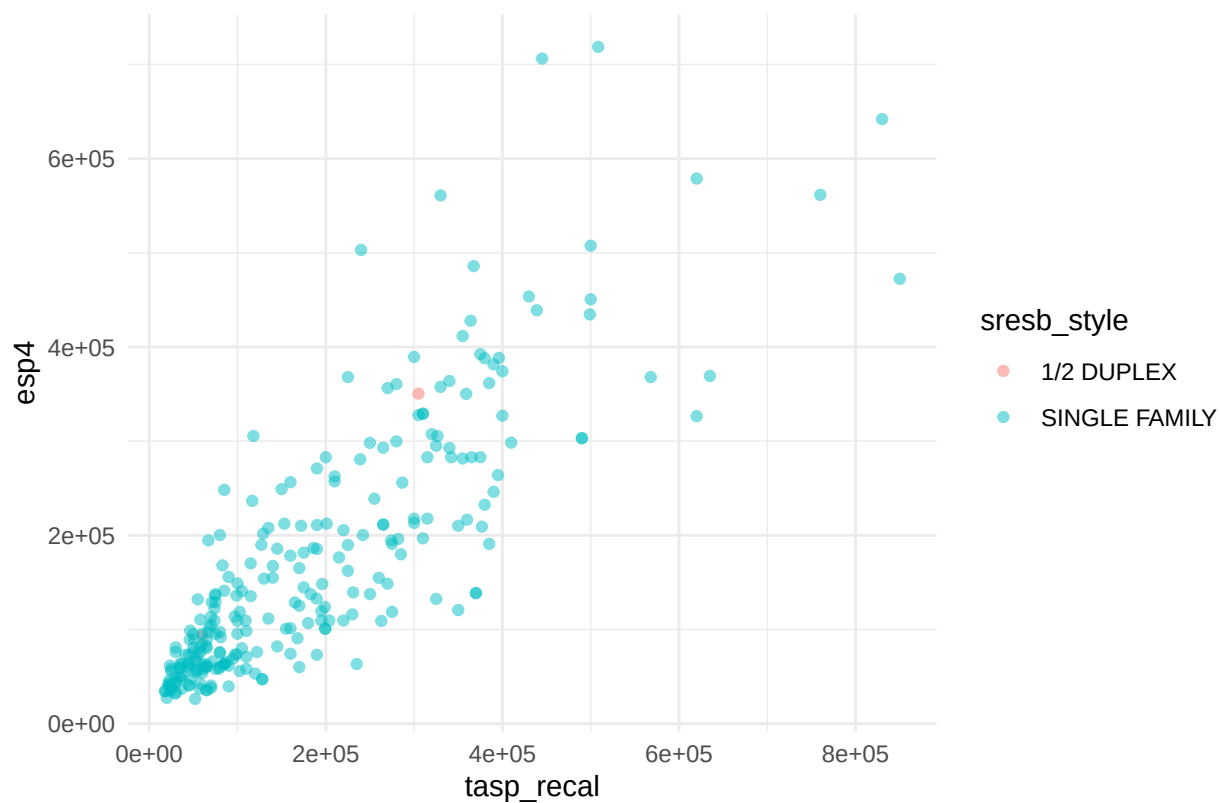
| metricsource       | df_total | df_resid | df_reg  | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|---------|--------|--------|--------|---------|
| SPSS               | 505      | 477      | 28      | 0.4578 | 0.7440 | 0.7290 | 49.5400 |
| R                  | 260      | 241      | 19      | 0.4851 | 0.7304 | 0.7092 | 34.3651 |
| % diff (R vs SPSS) | -48.51%  | -49.48%  | -32.14% | 5.96%  | -1.83% | -2.72% | -30.63% |

Table 10: District 5 | LN\_TASP Model II Assessing Statistics: SPSS vs R

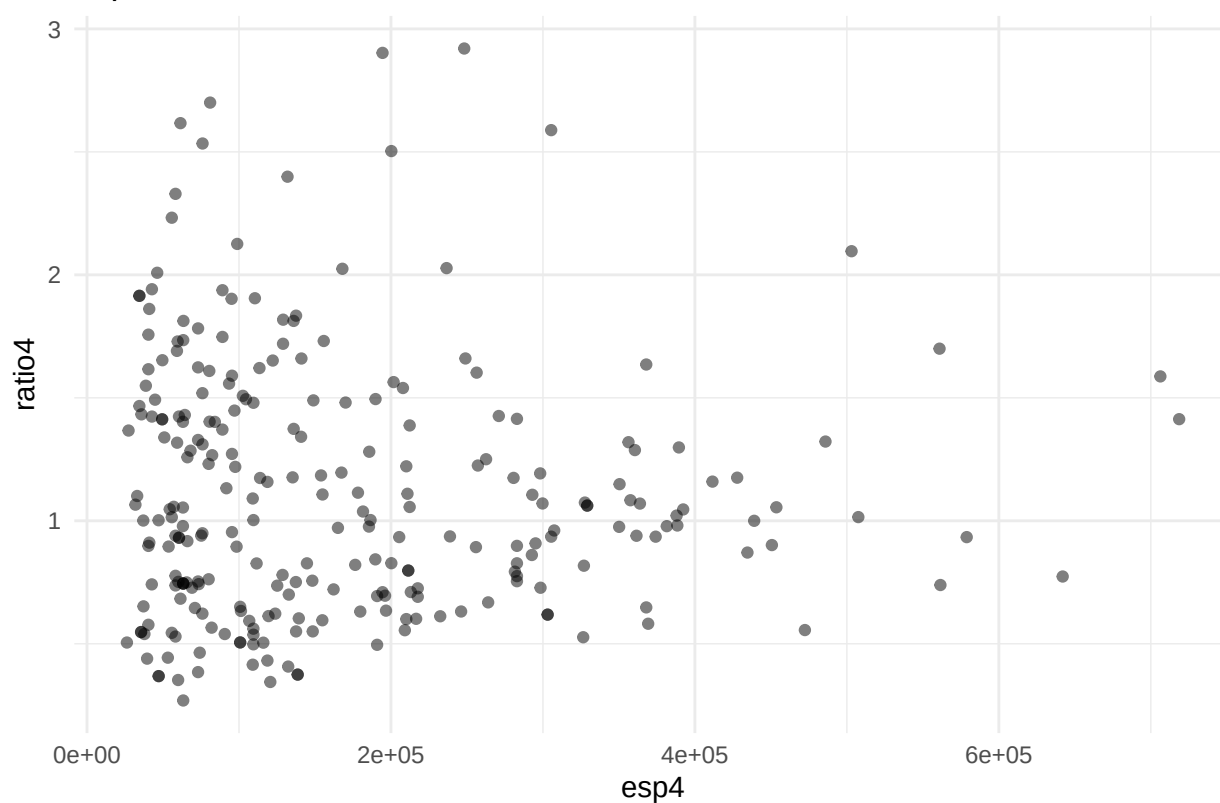
| metricsource       | median_ratio | mean_ratio | wmean_ratio | PRD    | COD    | PRB     |
|--------------------|--------------|------------|-------------|--------|--------|---------|
| SPSS               | 1.0000       | 1.1030     | 0.9290      | 1.1880 | 0.3790 | -0.1290 |
| R                  | 1.0030       | 1.1114     | 0.9270      | 1.1990 | 0.4016 | -0.1434 |
| % diff (R vs SPSS) | 0.30%        | 0.77%      | -0.22%      | 0.92%  | 5.97%  | -11.13% |

## Plots

tasp vs esp4 by Style



esp4 vs ratio4



Histogram of ratio4

